SECURE WEB DEVELOPMENT

SECURE CODING PRINCIPLES

CHARACTERISTICS OF SECURE ORGANIZATIONS

- Organizational management with security champions
- Proper information security policies
- Development methodologies with security checkpoints
- Secure release and configuration management

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DEVELOPMENT METHODOLOGY SELECTION

STRONG ACCEPTANCE OF DESIGN, TESTING, AND DOCUMENTATION

SECURITY CONTROLS CAN EASILY BE INTRODUCED

MATCHES ORGANIZATION'S SIZE AND MATURITY

REDUCES ERROR RATE AND IMPROVES PERFORMANCE

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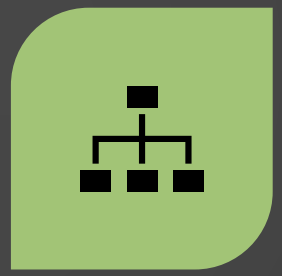

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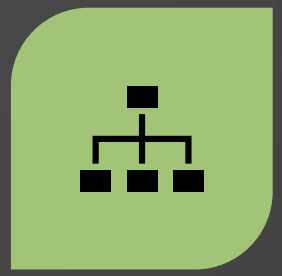

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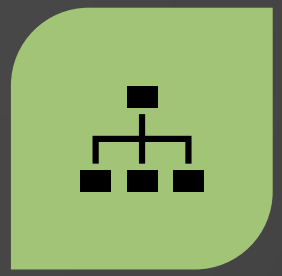

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REQUIREMENTS AND CODING STANDARDS

ARCHITECTURAL REQUIREMENTS

- Define how the system should be designed to meet security, performance, and compliance goals
- Key Aspects
  - Security – Enforce authentication, encryption, and secure communication
  - Scalability – Support increasing users and data without performance degradation
  - Modularity – Ensure components are loosely coupled for easier maintenance
  - Reliability – Prevent single points of failure through redundancy and fault tolerance
  - Compliance – Follow industry standards (e.g., GDPR, HIPAA, PCI DSS)

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DOCUMENTATION REQUIREMENTS

- Necessary guidelines, records, and reports that ensure security is integrated throughout the software development lifecycle (SDLC)
- Ensures transparency, compliance, and maintainability in software development
- Covers security policies, system design, coding standards, threat modeling, testing, and incident response
- Helps developers, auditors, and security teams understand and enforce security measures effectively

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TESTING REQUIREMENTS

- Security testing methodologies, tools, and best practices that ensure web applications have minimal vulnerabilities and comply with security standards
- Security testing is critical to identifying vulnerabilities before attackers do
- A combination of SAST, DAST, pen testing, and fuzz testing provides comprehensive coverage
- Authentication, access control, and secure configurations must be validated to prevent common attacks
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IN-CODE COMMENTING

- Practice of adding descriptive, security-aware comments within the source code to enhance readability, maintainability, and security awareness
- Helps developers understand security implementations, potential risks, and important security measures within the codebase

Explain security mechanisms to improve developer understanding

Warn about potential security risks in critical areas

Avoid storing secrets in comments to prevent leaks

Use comments to enforce secure coding guidelines and best practices

Keep comments accurate and updated to reflect security improvements

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EXCEPTION HANDLING

- Gracefully managing errors and unexpected conditions in an application while ensuring security, stability, and minimal exposure of sensitive information to attackers
- Proper exception handling helps prevent information leaks, denial-of-service (DoS) attacks, and unauthorized access

Hide detailed error messages from users to prevent information leaks

Use custom error pages instead of default web server error responses

Catch specific exceptions to avoid unintended behavior

Prevent crashes by validating inputs and handling unexpected errors

Log errors securely and monitor logs for security incidents

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NAMING CONVENTIONS

- Structured and consistent rules for naming variables, functions, files, and database elements to enhance code readability, maintainability, and security
- Proper naming conventions reduce security risks such as confusion, accidental exposure of sensitive data, and vulnerabilities from insecure identifiers

Avoid exposing sensitive information in names

Use consistent and meaningful naming conventions to improve maintainability

Prevent predictable identifiers in sessions, files, and API routes

Follow secure naming for database tables and environment variables

Never hardcode passwords, API keys, or sensitive data in variable or file names

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